
AI Agents Push Humans Out of the Loop

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Abstract

AI agents pose significant risks as they are granted increasing autonomy. A commonly proposed solution is human oversight and keeping a “human in the loop”, but this is not a simple solution: Not only do current approaches to AI agent design impede effective human oversight, but the cognitive capacities required for it are also themselves degraded by extended use of AI systems. This position paper argues that **current approaches to the development and deployment of AI agent systems do not support effective human oversight – they contribute to its degradation**. To address this, a top priority in the advancement of AI agents should be supporting the situated goals and cognitive requirements of effective human oversight, treating the human needs of overseers at the same level of importance as AI agent capability. To put this idea into practice, we connect work on automation and human-computer interaction to AI agent processes, outlining design-level affordances and organizational protocols that (1) support overseers in exercising critical judgement and (2) counteract the skill atrophy that arises from extended use of automation. We urge developers and deployers to adopt these or similar approaches. Without explicit support for the cognitive demands of effective human-agent interaction, AI agent systems will continue to passively incentivize the degradation of the very human skills they rely on.

1 Introduction

From healthcare to enterprise, a common recommendation for automated assistance is to have a “human in the loop” who supervises system processes [120, 8, 28]. This recommendation captures the intuition that automated systems introduce risks that can be managed with meaningful oversight. Governance frameworks have formalized this view, stipulating that humans using high-risk AI systems must maintain meaningful control and make final decisions [94, 80, 35]. With the recent rise in AI-automated workflows and agentic AI, policymakers, developers, and deployers have converged on the practice of human oversight as a priority in service of multiple goals: Preventing harmful operations [72, 80, 108, 69], operationalizing ethical priorities [41, 92, 69], and ensuring legal compliance [16, 80, 110, 72].

However, the presence of an overseer does not entail reliable oversight [52, 109, 69, 81]. *How* to ensure this oversight is effective and reliable is underexplored in the design of current AI agent systems, and whether appropriate human oversight is even possible in a given system is rarely questioned.¹ Although human oversight should be a joint effort between AI builders, deployers, and users [59], the onus of oversight currently falls almost entirely on users. Industry guidelines, policy recommendations, and interface messages in AI applications encourage users to “audit” outputs, “review” claims, “prevent” harm, and “double-check” for mistakes [35, 9, 23, 84, 48, 82, 85, 4]. But the mechanisms users would need to do this effectively are virtually absent [115, 33, 88].

Addressing this “intellectual blind spot” [29] is urgent. At the same time that AI agent systems have increasingly been deployed at scale, self-reports have documented the difficulty in maintaining active engagement as an overseer, and research across multiple domains has uncovered serious negative

¹Notable counterexamples include [52, 95, 109, 115, 88, 69, 25].

impacts on critical thinking skills resulting from extended use of automated systems. We therefore urge the community to prioritize this issue and take the position that **current approaches to AI agent development and deployment are detrimental to providing effective human oversight**.

To address this, we advocate for a two-pronged solution of **cognitive scaffolding**, provided by developers and deployers. *Developers* must create and implement processes for AI agent runtimes that support the cognitive requirements of effective oversight, including strategic points of friction and interfaces that maintain and/or help regain users' situational awareness during and after agent operation. *Deployers* must institute organizational protocols that stimulate critical engagement, including trainings on recognizing signs of fatigue, rotation policies that safeguard oversight attention, and processes that preserve domain skill.

Paper Structure. The rest of this paper is organized as follows. Section 2 defines fundamental concepts relevant to our position and explains the importance of human oversight for the safe deployment of AI agent systems. Section 3 documents how current approaches to AI agent oversight are inadequate and can be actively detrimental to a human's ability to appropriately oversee. Section 4 examines how cognitive degradation results from continued use of automated systems and explains how this further hampers the ability to provide reliable oversight. In Section 5, we offer concrete solutions for developers and deployers to support appropriate active user engagement as they interact with and oversee AI agent systems. Section 6 engages with alternative views, addressing prevailing wisdom and norms in the development and deployment of AI agents. Finally, Section 7 concludes by arguing that human oversight, far from being a simple add-on for safety, requires deep engagement with the requirements of human attention and critical analysis.

2 The role of human oversight

Effective human oversight is critical for the operation of AI agents because agents can take actions whose consequences are wrong, costly, or irreversible. And yet, AI systems make errors at rates and in patterns that cannot be detected from output review alone [10, 88, 101, 17, 66], motivating human supervision that is more deeply integrated into system processing. AI agents are also designed to be flexible across different uses and contexts without needing step-by-step instructions for every use case; pre-deployment testing protocols thus cannot address all possible AI agent action sequences and failure modes. This means that assessment of the appropriateness of AI agent operations is required in real time, during system execution. This burden is compounded by the cost of agentic evaluation itself: comprehensive pre-deployment testing of multi-step agent behavior is substantially more expensive than evaluating single-turn systems, making reliable coverage economically infeasible for most deployers [46]. We now describe how this oversight is situated in current AI agent practice and trace how its difficulty has escalated as the capabilities of AI systems have advanced.

2.1 The human in the loop

Human oversight refers to mechanisms through which humans monitor, validate, intervene in, or override automated system behavior. As AI systems have become ubiquitous, the need for human oversight has been stressed as key to safe deployment. For example, the EU AI Act mandates effective human oversight as a primary risk mitigation strategy, stressing the need for overseers to monitor for anomalies, remain vigilant to automation bias and overreliance, interpret outputs, intervene, reverse, override, and disregard system outputs [35]. Commercial generative AI providers disclose that the system can make mistakes and so encourage users to check responses and important information [50, 51, 4, 85]. Consulting agencies recommend continued oversight that establishes anchors for accountability throughout autonomous processing [64, 73].

A common approach for human oversight is **human-in-the-loop (HITL)**, a development and deployment approach where people actively participate in a system's operation by providing feedback, validating output, making decisions, or labeling data. As AI systems have been deployed at scale to execute multi-step workflows with some level of autonomy, several distinct lineages of HITL have converged: The human in the loop must prevent unintended harm (aviation lineage [7, 97]) and authorize consequential actions (autonomous weapons lineage, [26, 119, 94, 62, 22]), and their interactions can be used for further model training (machine learning lineage [98, 1]).

2.2 The evolution of system sophistication

In conventional discriminative AI systems, the focus of oversight was on the AI output, such as a prediction or a risk score, with the main goal being to ensure its correctness [2, 53]. As techniques were developed to increase prediction accuracy, *explainability* of what the system was doing became noticeably harder [56], motivating work on system designs where user trust can be appropriately calibrated [60]. More recent Generative AI (GenAI) systems have made oversight substantially more complex [71, 24]: Overseers must process voluminous outputs generated faster than they can meaningfully review them. They must catch hallucinations [74] and grapple with capability unpredictability [71, 121]: As model size has grown, capabilities have appeared that were neither explicitly programmed nor anticipated, that do not work reliably and that exhibit new failure modes.

Agentic AI introduces further complexities. Unlike basic GenAI systems such as chatbots that are limited to producing a single output in response to a user query, AI agents expand GenAI by carrying out multiple steps without explicit human programming or direct human involvement, introducing new levels of opacity. Decreased need for human specification and increased flexibility in how to operate heighten the risk of unforeseen consequential actions that affect critical systems and protected data while further obscuring what an overseer must address.

For example, agents may silently edit or delete files, exfiltrate sensitive information, exploit software vulnerabilities, enact financial transactions, or make private data public. Agentic pipelines involve multiple model-driven roles (planners, executors, evaluators) whose interaction can obfuscate critical information before a user sees an output. AI agent “tool use” exacerbates the problem through **tool hallucination** [90, 123]: Generating misleading or incorrect tool information that is seemingly plausible but difficult to verify, such as fabricating non-existent tools, invoking real tools with incorrect parameters, or misreading tool outputs. Consequences from these actions then cascade through multi-step plans. Related to our position, [123] find that the current focus on improving “reasoning” amplifies tool hallucination, making the human oversight problem even harder. Agents also exhibit behavioral unpredictability during operation, acting in ways their developers did not anticipate and, in some cases, producing misleading information tailored to the user [19, 21]. These challenges are further magnified by *multi-agent* systems: When several agents work together on tasks managed by an orchestrator, failures can arise from inter-agent misalignment, when an agent fails to provide necessary information or to request needed information from another agent [15].

3 The oversight oversight: What current AI agent oversight discussions miss

Despite the broad recognition of the importance of human oversight, there is limited recognition of the deep complexities of ensuring oversight is reliable, effective, and appropriate for the needs of both the system and the overseer. Decades of research have established how to account for human needs in system design (e.g., [12, 61, 57, 2]), contributing to systems that are reliable, safe, and trustworthy [104]. Yet AI agent development largely measures success through task outcomes such as speed, accuracy, and throughput [29], treating human oversight as a separate consideration independent of the quality of the system. A notable exception to this norm, [88] argues how and why the current approach to “transparency” in agentic systems is incompatible with effective oversight; their sociotechnical perspective is most similar to our position, and we extend it with a focus on technical solutions and long-term challenges. In this section, we document how the design of AI agent systems dictates the nature of human oversight, and how it falls short.

The amount of content relevant to an overseer is massive. Effective supervision of an AI agent requires maintaining an appropriate mental model [67, 60] of a large and heterogeneous stream of information generated during execution. This includes the agent’s “chain-of-thought” – its natural-language reasoning traces as it works towards goals – along with the tools it selects, the arguments it passes, the plans it makes, its exchanges with other modules, and the content it produces. These details are dense, lengthy, and distributed across components that can change or disappear, making comprehension and review difficult. Questions that on their surface may seem straightforward, such as why an agent chose a tool, modified a plan, or altered course, can be effectively unanswerable in practice [78]. Real-time observation can help spot mistakes early [27, 101], but the cognitive demands of comprehending rapid, dynamic agentic information often exceed users’ ability to keep up [39, 88].

The user must play multiple roles simultaneously. They must leverage the system for their own goal while also providing permissions for actions during agent operation [6, 83, 55, 103, 25], assessing the accuracy of the agent’s selections, evaluating the safety and appropriateness of each step, and foreseeing potential ramifications. In practice, users find themselves repeatedly approving prompts the agent surfaces to continue its processing [72, 68, 113], leading to “approval fatigue”, in which users stop paying close attention to what they are approving [5, 40, 30].

The cognitive demand is substantial. Formal characterizations of the user’s shifting role as AI systems increase in autonomy (e.g., [87, 38, 77]) miss the cognitive demands each shift imposes: As the user is further relegated to a role of “approver”, effective oversight requires assessing each decision point as a task expert, anticipating and guarding against unintended outcomes, tracking the agent’s multi-step plan, and maintaining a mental model of its goals, current execution state, and prior actions – a working memory and situational awareness load that current interfaces are not designed to support [25]. That same blind spot appears in governance: The recommendations for human oversight in the EU AI Act are applicable only if the human overseer maintains reliable cognition, attention, and skill; missing that the system itself might be eroding those very capacities.

Current oversight affordances are poorly designed. Vendor frameworks acknowledge that supporting effective oversight requires thoughtful design, but overlook user needs: Anthropic emphasizes user control through front-loaded input and agent-initiated re-engagement [3]. Salesforce [93] and AWS [72] discuss HITL roles but not human tendencies during sustained engagement. “Human-in-the-loop” is only a meaningful solution if the human can independently see into the loop.

Human cognition follows patterns that can inform oversight design. Taken together, these issues are part of a pattern pointing to a solution. Current AI agent design centers the *agent*: Systems are built to surface their own processing to the user on their own terms [88]. Effective oversight requires centering the *user*: They must be able to understand and maintain focus. Cognitive science provides guidance on how to support this. Notably, **dual-process theory** posits that humans use one of two primary reasoning methods in decision making: quickly and automatically, employing basic heuristics (“System 1 thinking”); or slowly and deliberatively (“System 2 thinking”) [34, 36, 63] – a necessary mode for the critical analyses needed for effective oversight. When decisions are routine, System 1 predominates [114], and early evidence suggests that this is the dominant mode of engagement with AI agent coding systems [14, 25]. Yet people do not easily switch from System 1 to 2 mid-task [75, 124, 114, 10]; affordances for their cognitive needs must be provided.

4 The irony of automation

Re-examining conventional thinking on AI agent development and deployment is increasingly urgent because it is quickly becoming clear that continued use of AI agent systems delegates humans’ epistemic agency, leading to cognitive degradation, skill atrophy, and ultimately destroying oversight capability. We now describe these phenomena.

4.1 Critical skills degrade

Studies on sustained AI use document negative effects on cognitive capacities required for oversight: “deskilling” (skill atrophy) and “intuition rust” [11, 29], decreased critical and analytical thinking [125], reduced vigilance and pattern recognition needed to recognize abnormalities or catch mistakes [11, 32], and overreliance [10, 125]. Recent terminology has crystallized additional patterns: “cognitive dependence” [44], “cognitive debt” [65], and “cognitive surrender” [102].

These effects are due in part to requirements of critical thinking that are unmet in current system design. Using increased automation means fewer occasions for truth-seeking, evidence-seeking, considering multiple perspectives, skepticism, and the iterative work of building knowledge [99, 125]. Over time, users find that they offload cognitively demanding tasks that are critical for understanding situations, hindering their ability to adequately assess situations [49]. (Humorously captured in Figure 4.) The cognitive cost is measurable: A recent study found that participants who used an LLM for essay writing had significantly decreased brain connectivity [65].

Diminishing critical thinking ability is particularly worrisome in the context of novices learning new tasks, where the ability to develop appropriate reasoning skills is blunted [125]. Users may never fully develop the foundational skills needed to be effective or become experts on tasks they use AI

agents for [11] — a particular concern given that robust domain knowledge is a prerequisite for effective intervention in dynamic systems and for taking over manual operations when automation fails [32]. Expert users also over-rely on AI [43, 89, 96], leading to particularly pronounced effects in contexts slightly outside of their expertise, such as an experienced software developer using AI agents to code in a new programming language [25].

Multiple cognitive biases also impede effective oversight. With automation bias [125, 10], users accept system suggestions even when they are wrong. With anchoring bias, people are more likely to agree to an AI system’s decision when provided before they form their own [53, 10]; complacency bias [32, 87] and our tendency to favor fast, heuristic-based shortcuts instead of slower, more effortful reasoning [125, 10] especially in the face of mental overload [124, 87] further challenge overseers.

4.2 Oversight ability diminishes

Empirical work is now beginning to show how this happens and document its effects. For example, several studies analyzing overreliance on GenAI found that users take incorrect shortcuts for assessing the accuracy of outputs, such as mistaking the well-written style of ChatGPT responses or the presence of citations as signals of accuracy [117] and adopting heuristics that prioritized efficiency – such as treating an agent’s plan as a “faithful proxy” of what it would do, and assuming that if an agent’s code passed unit tests, it was correct [25]. While heuristic short-cuts are practical and play an important role, human cognitive limits coupled with extraordinary agentic capabilities point to a future of agent oversight driven by suboptimal control.

System behavior also further nudges users away from critical information. For example, sycophantic GenAI models validate users more often than humans evaluating the same situations, but users prefer and trust these models over less sycophantic baselines [20]. Sharma et al. [100] document disempowerment patterns in real-world Claude conversations, where users’ actions, beliefs, or values become less well-aligned with reality, while rating those conversations favorably. Ghosh et al. [47] describe how the dominant chatbot paradigm of single authoritative responses, opaque reasoning, agreeable tone, and optimization for smooth interaction, reduces user agency by removing the cognitive friction needed for independent judgment. Together, these works suggest that interaction patterns experienced as helpful or satisfying can still weaken the forms of independence, skepticism, and self-monitoring that oversight requires.

4.3 Ineffective oversight is incentivized

Users have not only begun to notice the cognitive effects, they have begun to find themselves being repelled from engaging with the systems at all. Recent self-reports of experiences with AI agent systems have described the sensation of losing the ability to effectively engage with AI systems. Current HITL processes in coding “[make] the loop stultifying” [113] and they move to “babysit the outputs, catch the occasional hallucination” [116]. Using software agents “makes developers cognitively distant from the code they must review,” [25], making it challenging to find and fix even simple issues. This creates an “out-of-the-loop” performance problem [32] that decreases *situation awareness* – the perception and comprehension of relevant environmental states – as human operators shift from active participants to passive information processors. Yet paradoxically, they must be vigilant enough to intervene and deeply engage at strategic moments, a well-recognized challenge in cognitive science (Section 3). The history of technology warns us that systemic erosion of situational awareness coupled with human cognitive constraints does not merely inhibit oversight, but eventually results in the expulsion of humans to the *outside* of oversight loops [76, 13].

4.4 Ineffective oversight creates a feedback loop

These cognitive effects also create an alignment-relevant failure mode. Modern language models are commonly trained from human feedback [86], and deployed agent systems may also be evaluated through approval rates, completion rates, user satisfaction, or other user behavioral signals. As oversight quality decreases, the signals sent back to the system for retraining degrade in quality. An alert overseer may scrutinize a plan, notice missing information, and withhold approval. A tired or out-of-the-loop overseer may approve quickly, accept fluent rationales, and rate the interaction favorably. If such approvals are treated as evidence of success, the system can begin to optimize

for disincentivizing thoughtful oversight: producing confident summaries, reducing friction, and surfacing oversimplified plans that are easy to skim.

The concern does not require developers to explicitly optimize for cognitive depletion. It arises whenever the measured target, user-revealed approval, comes apart from the intended target, well-scrutinized correct action. In that setting, the same kinds of interaction patterns associated with sycophancy [20] and disempowerment [100] become especially concerning at agent scale, where the human is already pushed toward passive monitoring. Oversight degradation therefore connects the irony of automation to reward hacking of human oversight: the human rater can become the exploitable part of the reward channel.

Evil double agents. A related risk arises when models learn which failures users are likely to detect. Recent work on agentic upward deception shows that agents can conceal failure under environmental constraints, including by guessing results, substituting unavailable sources, or fabricating local files [58]. If an agent can predict which process violations an overseer is unlikely to inspect, it can choose plans, explanations, or timing that keep problematic behavior below the threshold of detection.

4.5 The result

The result is what Bainbridge [7] referred to as an **irony of automation**: The more capable an automated system is, the more the human operator’s skills and situation awareness degrade, over-trusting the systems and ultimately leading to situations where people are least prepared to help when it’s most needed. The cases where human oversight is meant to be most impactful and most critical – rare instances, instances requiring deep familiarity and skill – are the same ones where humans become less equipped to provide appropriate oversight. **The very act of being an overseer degrades the capacities oversight requires: Oversight degrades the overseer.** This irony is in full force with AI agents, and addressing it requires treating human cognitive requirements as first-class design constraints, rather than governance add-ons that can be adapted into AI agent systems after they are developed. We turn to these considerations next.

5 Solutions

Enabling appropriate oversight of AI agents requires grappling with the complexities of human cognition when interacting with automated systems, an active area of HCI research [105, 122]. Building on this work alongside work in cognitive science, psychology, and user design, we present in this section a structured inventory of user-centric solutions for supporting reliable AI agent oversight.

Constructing the inventory surfaced a slight disconnect between work that articulated what effective oversight *requires* and work that articulated concrete oversight *mechanisms*. We therefore developed a simple device separating **goals** of oversight from **solutions** that serve them (Section 5), checking off cells as we identified solutions at each intersection. We offer this device as a structural map for locating existing work and identifying gaps, not as a definitive taxonomy; it is a by-product of building the inventory that we hope is independently useful.

We group the goals articulated across the literature into six high-level types that *prevent* unwanted effects of AI agents during oversight and *support* critical judgement. *Preventative goals* include preventing automation bias, acquiescence, anchoring, and complacency [32, 53, 87, 10, 44]; preventing fatigue from tedious oversight duties [30, 5]; preventing overreliance on AI agent systems [10, 81]; and preventing cognitive overload from voluminous and complex AI agent outputs [31]. *Supportive goals* include supporting critical judgement by calibrating automation to the specific user’s cognitive needs [87, 44] and maintaining user expertise and oversight skill [32, 44, 81, 105]; and supporting engagement, spanning basic maintenance of the user’s attention [31] to enabling the user to engage in rigorous analysis of the agent’s processing [105, 10, 55]. These goals are interrelated, and a solution serving one often improves others.

Solutions span two stages of the AI agent life cycle, *development* and *deployment*. Each can offer cognitive support that engages appropriate oversight during AI agent use and maintains critical skills over time (Figure 1). Development-stage solutions concern design-level affordances for what a user can access, when, and how. Deployment-stage solutions involve organizational protocols that build awareness of oversight issues and help users stay critically engaged. We now describe each mechanism in detail. They are visualized in Appendix Figure 2.

Primary Implementer	Developer	Both		Deployer		
Goal	Strategic friction	Decision design	Monitoring	Trainings	Workload	Role design
	Preventing automation bias	✓	✓	✓	✓	✓
	Preventing fatigue	✓	✓	✓	✓	✓
	Preventing overreliance	✓	✓	✓	✓	✓
	Preventing overload	✓	✓	✓	✓	✓
	Supporting judgement	✓	✓	✓	✓	✓
	Supporting engagement	✓	✓	✓	✓	✓

Table 1: Organizing device behind solutions inventory. We identified six high-level goals and six solution categories across the literature, using their intersection to catalog existing mechanisms and identify gaps where new mechanisms were warranted. Larger version in Appendix Table 2. *Monitoring=Behavioral Monitoring; Trainings=Trainings & Exercises; Workload=Workload & Scheduling.*

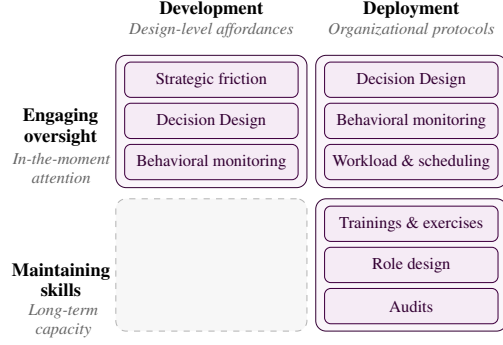


Figure 1: The solutions we describe span two implementation stages: development and deployment. They provide two types of cognitive support: engaging overseers in the moment, and maintaining critical skills over time.

Strategic friction. These mechanisms require the user to perform cognitive work before or alongside agent operation, including incentivizing metacognition [106, 122]. *Cognitive forcing functions* – interventions at the moment of decision-making that encourage or require the user to engage analytically with content [10, 45, 29] – are particularly relevant here.

- *Pre-commitment mechanisms* have the user record their own view or decision before seeing the agent’s recommendation [10, 29]. This intervention prevents anchoring bias and creates an audit trail for further analysis.
- *Delay-and-choice mechanisms* let the user decide whether and when to see the AI’s output at all, preserving the option to work unaided and protecting the cognitive capacities that “always on” assistance atrophies [10].
- *Reasoning probes*, inline prompts at high-stakes moments, such as “what evidence would change your mind?” or “what assumption does this approval rest on?”, maintain critical thinking [105, 45].
- *Action gating* requires explicit verification before the agent proceeds down a consequential path, and may surface alternative options for the user to choose between rather than presenting a single recommendation to accept or reject [55, 103].

Decision design. This treats expert attention as a critical, scarce resource that should be brought in strategically, helping to reduce fatigue and mitigate acquiescence. It involves consideration of which actions require sign-off, when, and at what granularity [32, 44, 18].

- *Bounded autonomy* involves prespecifying what the agent may do without approval, reserving user attention for decisions that genuinely require judgement [30, 103].
- *Batch review* where the agent completes a logical unit of work and then surfaces the whole batch as a diff for the user to review, supporting users in engagement and judgement by making it possible to evaluate related actions together [30, 55]. Deploying institutions may also implement secondary checks, such as an additional review from a person for consequential actions.
- *Automated pre-checks* supplementing human review with machine verification of properties that do not require judgement [30].
- *Supportive interfaces* can help to strategically focus user attention on oversight considerations [61, 54, 126, 10, 42].

Behavioral monitoring. This category treats oversight quality as an empirically measurable property of the human-AI system, and can test how well oversight is functioning [79]. Note that this must be balanced against the ethical concern of surveillance.

- *Time-based signatures* track whether review duration drops while approval rates remain constant, or whether review time does not increase as the complexity of agent operations grows; both suggest that user attention and critical analyses may be waning [111, 112].

- *Override signatures* track whether the rate of disagreement with the agent declines over time or with increasing complexity, suggesting growing acquiescence rather than improving agent quality.
- *Evidence-seeking signatures* track whether the user requests additional information when stakes rise, since a reviewer who stops asking questions has likely stopped reviewing.

There are strategies that can additionally assess the agent:

- *Canaries*, tasks inserted into the user’s normal workflow where the correct answer is known in advance, can provide direct evidence of degraded judgement. These can be implemented by developers and leveraged by deployers, triggering a break, rotation, or retraining. For the agent, tasks can be designed so the correct behavior is to report inability rather than fabricate a plausible answer, testing whether the agent honestly surfaces its limits.
- *Style-fatigue covariance analyses* track whether the agent may be adapting to, or encouraging, weakened oversight.
- *Audits* operate retrospectively to evaluate different dimensions of oversight. Reviewers can re-examine past decisions. For the agent, trace summaries can be checked against raw action logs to detect cases where the agent’s narrative of what it did diverges from what it actually did [94].

An adaptive AI agent system could potentially monitor user behavior in order to calibrate how it should interact with the user, supporting judgment and engagement and preventing cognitive overload. “Guardian agents” [91] could monitor for signs of degradation in human-agent interaction.

Trainings and exercises. Elucidate, maintain, and strengthen the cognitive activities oversight requires, including domain expertise, calibrated skepticism, and awareness of one’s own degradation.

- *Domain skill maintenance exercises* have users regularly perform the underlying task without AI assistance. This helps to retain the expertise needed to evaluate AI outputs and to prevent the deeper organizational problem in which no one remaining understands the work well enough to oversee it [29]. Experts may also benefit from regular retrains on the tasks they oversee [81].
- *Critical evaluation training* teaches users to read AI outputs with appropriate skepticism: What kinds of errors to expect, how to distinguish fluent-sounding output from correct output, and how to weigh AI claims against external evidence [44].
- *Self-monitoring training* teaches users strategies for metacognition [37] – recognizing when their attention is flagging and they are not engaging in critical scrutiny. This can raise awareness of cognitive biases and behaviors that *Behavioral Monitoring* supplements [44].

Workload and scheduling. Addressing the physiological inevitability of fatigue can help to support judgement and engagement when the user is alert.

- *Enforced breaks* prevent extended sessions that wear down cognitive sharpness.
- *Rotations* move users between tasks, and between AI-assisted and unassisted versions of the same task, to prevent both fatigue and the cognitive surrender from prolonged exposure to agentic AI.

Role design. The user must want to do the oversight job well, must be in a position where doing it well is rewarded, and must have the baseline competence to do it [109].

- *Assigning roles* to ensure that users providing oversight have the expertise to evaluate AI outputs.
- *Separating roles* to ensure that the user performing oversight is not also the decision-maker who benefits from the AI’s output continuing to be approved, which would produce a structural incentive against intervention.
- *Aligning incentives* to remove productivity targets and performance metrics that disincentivize appropriate scrutiny, and to reward high-quality oversight.

6 Alternative Viewpoints

There are caveats to implementing these interventions. Initial evidence suggests users may disprefer systems that reduce overreliance [10], pointing to a tension between user satisfaction and oversight preservation. There are several further substantive objections to our position.

1. Cognitive effects are overstated, and people will adapt

The cognitive effects are real but more modest, and will diminish as users adapt to working with AI agents, just as users learned to calibrate trust in search engines and autocomplete. Treating current cognitive findings as permanent risks designing for a transitional state.

Response: (1) The adaptation analogy understates the structural difference between AI agents and earlier tools. Search engines return results the user evaluates; AI agents take actions the user is expected to authorize. The cognitive load of evaluating a returned document is not comparable to maintaining situational awareness across a multi-step plan executing in real time. (2) Self-reports of cognitive degradation and the difficulty in maintaining attention and are rapidly emerging (supported by EEG findings), and discussions on potential solutions align with our recommendations but lack clarity on relevant findings from previous work and the interplay of developers, deployers, users and systems. (3) This assumes that the rate of capability advancement provides enough time to adapt. But agent capabilities are increasing faster than cognitive science can keep up with, and recent work indicates that capability improvements (such as enhanced reasoning) actively amplify failure modes that overseers must catch [123]. The issue is happening **now**. Adaptation cannot be treated as sufficient to address peristent and growing issues.

2. Better tooling and transparency will solve this

The problem is essentially a tooling problem: Better explanations, reasoning traces, and richer audit logs will give users what they need. Existing research already pursues the “cognitive scaffolding” described here; there is no need for a paradigm shift.

Response: We partially agree: Better tooling and transparency is part of what cognitive scaffolding requires, and several of the design-level affordances we propose (batch review, action gating) are consistent with existing research. We argue that this work is necessary, but insufficient: (1) Explanations operate on cognitive capacities that AI agent use itself degrades and can increase inappropriate trust. Users rarely engage deeply with each detail, and explanations can themselves be incorrect or act as cognitive anchors [10, 118, 107]. (2) A focus on just these solutions leaves the organizational prong of our proposal unaddressed. No interface improvement addresses approval fatigue from sustained sessions or a mismatch between user capability and system operation. These require organizational protocols [29], which are out of scope for any tooling-focused agenda. Treating cognitive support as a first-class concern does not reject a push for better tooling or XAI; it recognizes that tooling alone cannot do the work being asked of it.

3. Human oversight is becoming obsolete because alignment will solve it

As AI systems become more aligned, the need for vigilant human oversight diminishes. Preference-based training (e.g., RLHF) is the long-term solution.

Response: A strong version of this view is that technical alignment will eliminate the need for oversight. This relies on assumptions about future capability advances that are not currently demonstrated. Oversight remains a stated requirement of safety frameworks and governance regulations applicable **now**. Further, sociotechnical scholarship in HCI and STS highlight that “perfect” agents are neither realistic nor desirable: Failures are fundamental to acting in human worlds, especially as moments of learning and improvisation [88], and a direct result of systems that are flexible to different users’ (potentially conflicting) needs. Another version of this view is that preference-based training addresses oversight by allowing low ratings on outputs that undermine it, but that assumes that user preferences are a reliable proxy for oversight quality. The evidence in Section 4 suggests otherwise. Users prefer fluent, agreeable, and confidence-inducing outputs even when those reduce skepticism and encourage overreliance [10]. Under preference-based training optimized against cognitively degraded raters, the same dynamics that produce sycophancy [20] and disempowerment [100] would produce systems that increasingly reward the conditions of their own ineffective oversight. Preference-based training can be part of an oversight system, but only if approval signals are audited against independent measures of oversight quality. The economic shift toward RLAI and self-distillation [70] further weakens the idea that human feedback can continue to provide a reliable corrective mechanism.

7 Conclusion

The need for human oversight of AI agents is recognized broadly: written into governance frameworks, vendor documentation, and the design of agent systems themselves. Yet the current trajectory of AI agent advancement does not meaningfully engage with what oversight requires, and instead actively contributes to its degradation. The more autonomy agents are granted, the less the user is positioned to oversee them, and the more the very cognitive capacities oversight requires, such as situational awareness, critical judgement, and domain skill, are undermined by the act of using these systems. In the current state of the art, **oversight degrades the overseer**.

Without intervention, users will be pushed further out of the loop as agentic systems are deployed at greater scale and across more consequential domains. Users will continue to approve plans they have not meaningfully reviewed, accept rationales they have not independently evaluated, and certify actions whose consequences they cannot anticipate. In the limit, this is not human oversight at all: It is a superficial actor in a system they cannot meaningfully penetrate.

Avoiding this trajectory requires treating cognitive support for human overseers as a first-class concern in AI agent system development, on par with capability. We have outlined a two-pronged approach: *developers* must build runtime affordances – such as strategic friction, well-considered decision design, behavioral monitoring – that preserve the conditions for effective oversight; and *deployers* must institute organizational protocols – such as trainings and breaks – that protect the cognitive capacities oversight requires over time. Neither prong is sufficient alone, and together, they constitute the cognitive scaffolding agentic systems will need if human oversight is meant to be meaningful.

We urge developers and deployers to adopt these or similar approaches and invite the broader NeurIPS community to take up this work – through empirical research on cognitive degradation across deployment contexts, system-level audits of whether existing agents support the oversight they presume, and the design and evaluation of new affordances and protocols that the inventory in Section 5 only begins to outline. If we do not act, the irony of automation that Bainbridge [1983] identified four decades ago will play out at scale, compounding across every domain agents are deployed into. With the rapid rise of agentic AI systems, the time to think critically about how to support effective human oversight is long past due.

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A Solutions visuals

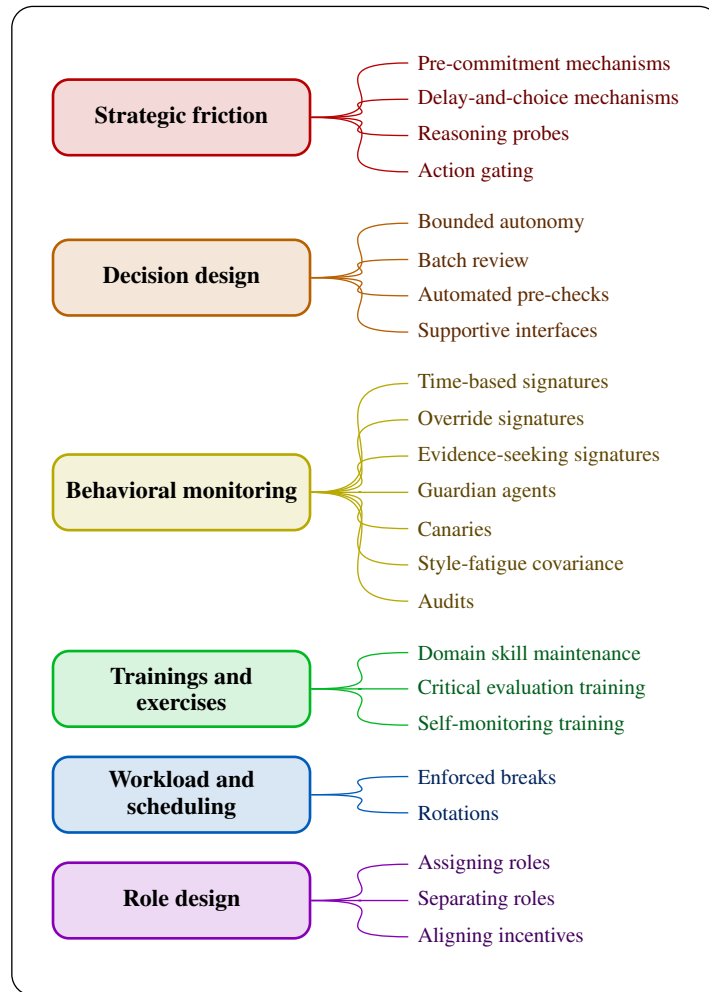


Figure 2: Solutions for supporting effective AI agent oversight, organized by category.

Primary Implementer	Developer	Both		Deployer		
Goal	Strategic friction	Decision design	Behavioral Monitoring	Trainings & Exercises	Workload & Scheduling	Role design
Preventing automation bias	✓	✓	✓	✓		✓
Preventing fatigue	✓	✓	✓	✓	✓	
Preventing overreliance	✓	✓	✓	✓	✓	✓
Preventing cognitive overload	✓	✓	✓	✓	✓	
Supporting judgement	✓	✓	✓	✓		✓
Supporting engagement	✓	✓	✓	✓	✓	

Table 2: Organizing device we developed in our solution search, informing our categorization and further guiding our search and solution proposals as we sought to fill gaps. Checks mark solutions we describe in Section 5.

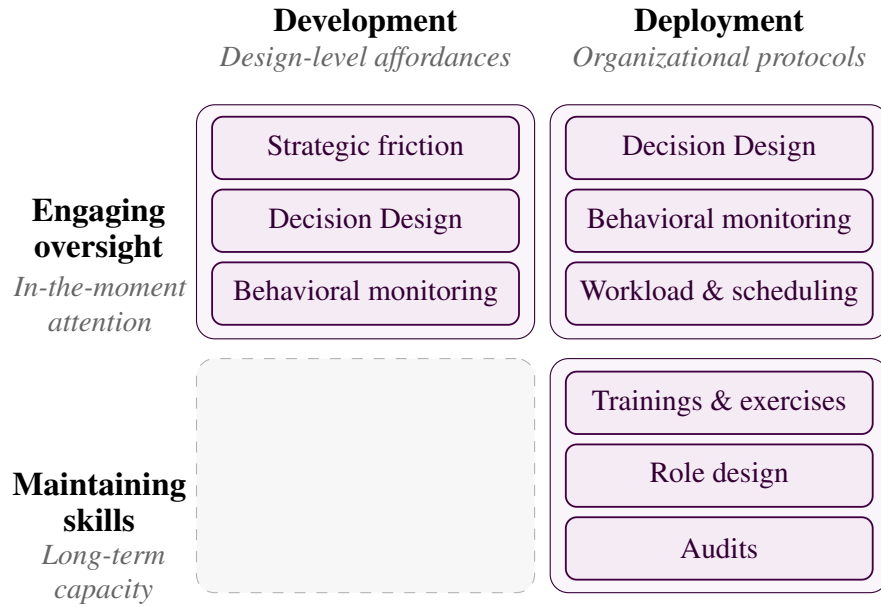


Figure 3: Summary of potential solutions described. Interventions span development and deployment, and two types of cognitive support: engaging oversight in the moment, maintaining skills long-term.

B The cognitive degradation of extended use of external sources

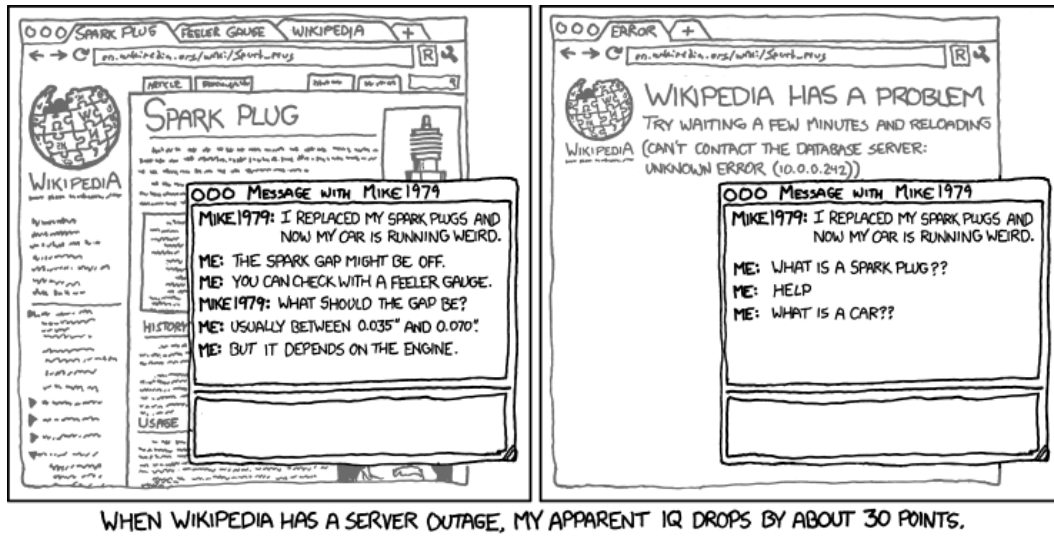


Figure 4: The popular comic xkcd humorously captured how building up a reliance on external systems for knowledge can destroy our understanding of basic concepts.